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Consensus Statment

Severe adverse events associated with general anesthesia with halogenated agents in patients of Venezuelan ancestry. Recommendations for clinical practice

Eventos adversos graves asociados a anestesia general con agentes halogenados en pacientes de ascendencia venezolana. Recomendaciones en práctica clínica

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ABSTRACT

The mtND4 Working Group of the Spanish Society of Anesthesiology and Resuscitation (SEDAR) has issued clinical practice recommendations. This article stems from reports of neurological injury and death following general in adult and patients of maternal Venezuelan ancestry, in whom a possible mitochondrial susceptibility associated with the mtND4 m.11232T>C variant of respiratory chain complex I has been identified.

This document is for informational and guidance purposes only. The final clinical decision should be individualized for each patient, taking into account the urgency of the procedure, therapeutic alternatives, and the resources available at each center.

RESUMEN

El grupo de trabajo mtND4 de la Sociedad Española de Anestesiología y Reanimación (SEDAR) emite unas recomendaciones de práctica clínica. Este artículo surge a partir de la comunicación de casos de daño neurológico y muerte tras anestesia general en pacientes adultos y pediátricos de ascendencia venezolana por línea materna, en los que se ha identificado una posible susceptibilidad mitocondrial asociada a la variante mtND4 m.11232T>C del complejo I de la cadena respiratoria.

Este documento tiene carácter informativo y orientativo. La decisión clínica final debe individualizarse en cada paciente, teniendo en cuenta la urgencia del procedimiento, las alternativas terapéuticas y los recursos disponibles en cada centro.

Palabras clave:

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Current situation

Several cases have recently been described in South America, Europe, and the United States of previously healthy patients or series of patients with Venezuelan maternal ancestry who have presented with infarcts of the basal ganglia of the brain, severe neurological deterioration, and in some cases, death after undergoing general anaesthesia.^{1–11}

In a subgroup of patients, investigators have identified a variant in the mitochondrial ND4 (mtND4 m.11232T>C) gene, a subunit of complex 1, that increases sensitivity to halogenated anaesthetics and contributes to neurological vulnerability.^{1–4,11} On 27 January 2026, the American Society of Anesthesiologists (ASA) and the Society for Pediatric Anesthesiology (SPA) issued a joint statement that includes several provisional recommendations that SEDAR included in its 2025 report. This document updates these recommendations.^{1,2}

Population at risk

All fully documented cases were of maternal Venezuelan ancestry and were asymptomatic before exposure to the anaesthetics.^{1–4,11} Since mitochondrial DNA is inherited exclusively maternally, any patient with a direct Venezuelan maternal lineage is considered at risk, even if there is no family history of anaesthesia-related complications.^{1–4,11}

Some of the patients described have a familial history of serious events or death following general anaesthesia in maternal relatives, which reinforces the suspicion of a mitochondrial aggregation pattern.^{1,2}

Agents involved and pathophysiological considerations

Most of the affected patients had received sevoflurane for anaesthesia maintenance, although full information on concentrations, duration, and systematic monitoring of anaesthetic depth is not available.^{1–4,11} Some of these patients subsequently tolerated intravenous propofol anaesthesia without apparent incident, although the safety of prolonged infusions is unknown.^{1–4,11}

Complex I of the electron transport chain is a known target of halogenated anaesthetics, and inhibition of this enzyme can reduce ATP production and cause metabolic ischaemic lesions in vulnerable structures such as the basal ganglia.^{1–3} Other anaesthetics, such as propofol or barbiturates, can also affect the electron transport chain. Midazolam, dexmedetomidine, ketamine, and short and ultra-short-acting opioids have so far not been implicated in these events.^{1–3}

Provisional recommendations

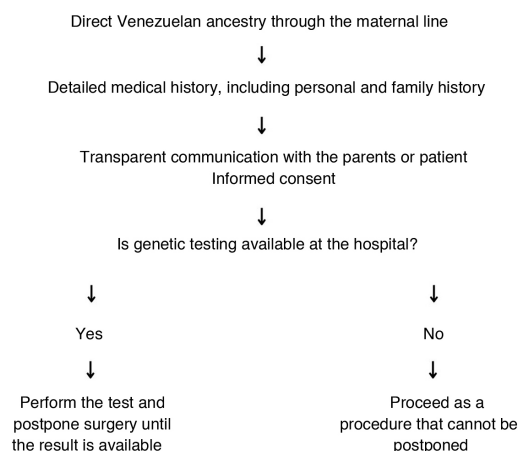
Preoperative evaluation. Risk screening

We recommend asking the patient specifically and respectfully if they have maternal Venezuelan ancestry (also consider the possibility of unknown maternal ancestry or egg donation), explaining the reason for the question and its relationship to a possible mitochondrial risk. All patients with direct Venezuelan maternal lineage should be considered at risk, regardless of whether or not there is a family history of anaesthesia-related complications.^{1–4,11}

It is advisable to investigate family history of:

- Significantly delayed emergence.
- Severe neurological deterioration, stroke, or basal ganglia lesions following general anaesthesia.^{1,2,5}
- Unexpected perioperative deaths.^{1,2}

Can the anaesthetic procedure be postponed?



Anaesthetic procedure cannot be postponed

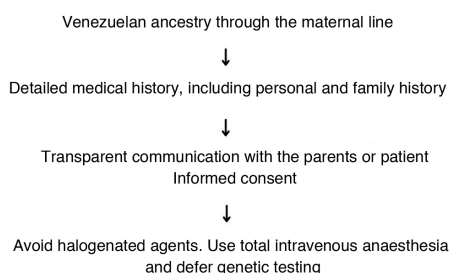


Figure 1. Clinical practice management algorithm: PREOPERATIVE.

Genetic testing

We recommend asking the clinical genetics department, if available, to perform mitochondrial DNA sequencing of the patient.^{1–4,11} It is essential to ask the laboratory to specifically search for the mtND4 m.11232T>C variant and to explicitly report its presence or absence. The genetics laboratory or department will run the test using either a blood sample or mucosal swab (Figs. 1–3)

Specific informed consent must be obtained for any genetic testing. The results must be interpreted by an expert in genetic diseases and family guidance should be made available.^{1,4,11}

Perioperative and anaesthesia management in patients of Venezuelan descent

The decision to postpone or proceed with the intervention should be based on the urgency of the procedure, the anaesthesia-surgical risk, and the available alternatives.^{1–4,11}

We suggest taking the following precautionary measures if the procedure is urgent or if genetic testing is not possible:

- Avoid halogenated anaesthetics (sevoflurane, desflurane, isoflurane), at least until more solid evidence is available.^{1–11} Purge the anaesthesia machine and circuits to remove traces of halogenated anaesthetics (cycle the ventilator, change tubing and soda lime, remove the vapor-

Anaesthetic considerations

- Consider **locoregional anaesthesia** with or without sedation, if possible
- **Anaesthesia machine**: purge to remove residual halogenated anaesthetic, including ventilator cycling, replacement of tubing and soda lime, and removal of the vaporiser
- **Fluid therapy**: prioritise balanced solutions, avoid Ringer lactate, minimise fasting, and avoid stressful situations
- **Premedication**: anxiolysis, EMLA, and venepuncture under nitrous oxide
- **Avoid halogenated agents**
- **Propofol**: meticulous titration. Use caution with high doses and prolonged infusions. Combine with remifentanyl and dexmedetomidine to reduce total doses
- Midazolam, dexmedetomidine, and ketamine are considered safe
- Fentanyl and remifentanyl: prioritise ultra short acting opioids

Monitoring

- Advanced haemodynamic monitoring: arterial catheter and frontal NIRS for long or complex procedures
- Monitoring of anaesthetic depth and cerebral function
- Monitoring of neuromuscular blockade: rocuronium and sugammadex
- Ventilation and EtCO₂ monitoring. Fresh gas flow equal to minute volume
- Intraoperative measurements, if available:
 - Lactate
 - pH
 - Glucose
 - Base excess
 - Anion gap
- Avoid leaving tourniquets in place
- Maintain parameters within the normal range, with particular attention to maintaining normothermia

Figure 2. Clinical practice management algorithm: ANAESTHETIC MANAGEMENT.

izer, consider using specific filters that reduce purging time if surgery is urgent).^{1,2,9}

- Prioritize regional (neuraxial or peripheral) or local anaesthesia techniques when indicated and feasible, combined with adequate sedation.^{1,4,9–11}
- If general anaesthesia is required, use total intravenous anaesthesia (TIVA) based on carefully titrated propofol, using combinations of midazolam, dexmedetomidine, ketamine, and short- or ultra-short-acting opioids. The use of propofol should be individualized, avoiding high doses and prolonged infusions.^{1–4,11}
- Monitor anaesthesia depth using processed EEG to avoid burst suppression and detect a possible increase in sensitivity to the agents used.^{1–4,8–11}
- Closely monitor normoxia, normocapnia, normothermia and haemodynamic stability, avoiding hypotension and cerebral hypoperfusion (consider the use of infrared spectroscopy, NIRS).^{1,2}
- Monitor blood glucose and acid-base status when mitochondrial dysfunction is suspected.^{1,2,7}

Postoperative monitoring

Maintain the standard outpatient surgery (MOS), admission or intensive care (ICU) patient pathway. Patients with a confirmed mutation or with neurological warning signs should be closely monitored in a postoperative monitoring unit.

In simple procedures, the absence of a paediatric intensive care unit is not in itself a criterion for referring the patient to another hospital. Tests should be performed to confirm that the patient's neurological

Postoperative care

- Consider admission to a postoperative high dependency unit according to the patient's age and availability:
 - Long and complex procedures
 - Delayed awakening or signs of neurotoxicity
- For patients requiring critical care, maintain intensive monitoring during the first 12 to 24 hours:
 - Neurological status and recovery
 - Metabolic parameters
 - Optimal pain control, avoiding stress and agitation
- If the patient's progress is favourable, consider transfer to a ward according to the established pathway.
- If surgery or examination under anaesthesia is performed as a planned ambulatory procedure, this pathway may be followed, except where there are findings suggestive of neurotoxicity.
- If focal neurological signs or altered consciousness develop, consider the following according to each centre's resources:
 - Activation of the stroke code protocol
 - Neuroradiological assessment, CT or MRI
 - Treatments agreed in consultation with Neurology

Notification

- Report adverse events through pharmacovigilance systems and the relevant national registries.

Figure 3. Clinical practice management algorithm: POSTOPERATIVE.

status has returned to baseline levels (level of consciousness, language, motor strength and focality).^{1–3,7}

In the event of neurological changes (abnormal emergence, seizures, focal deficits or sustained decreased alertness) local protocols for urgent neurological care should be activated (neurological examination, stroke protocol, neuroimaging tests, etc.) and metabolic monitoring should be started.^{1,2,7}

Notification and updates

Anaesthesiologists are urged to systematically report serious adverse events associated with this condition on their hospital's patient safety and pharmacovigilance systems, and notify all relevant national registries. Collecting clinical, anaesthesia, and genetic data is a key step in defining the magnitude of the risk, the frequency of the mtND4 m.11232T>C variant, and the safety profile of the different anaesthetic procedures.

SEDAR will continue to work with other scientific societies and groups of experts in mitochondrial diseases to review the evidence and update these provisional recommendations when new relevant information becomes available.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.redare.2026.502167>.

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